

A finite pointwise bound for alternating-block dice

Research notes, 30 September 2026

Status and attribution. The construction below is a dice realization of the classical shelf shuffle, up to reversing labels or inverting the permutation. Its pointwise mixing scale is already established by Diaconis, Fulman, and Holmes, *Analysis of casino shelf shuffling machines*, Annals of Applied Probability 23 (2013), 1692–1720, arXiv:1107.2961, Theorem 3.4. The purpose here is a self-contained finite-parameter estimate suitable for the fair-dice article. It improves that article’s stated approximate construction; it does not solve the exact-fairness open problems.

For $n \geq 2$, let

$$B_n = 12 \cdots n n \cdots 21, \quad s_{n,m} = B_n^m.$$

Each die has $2m$ faces. Let $P_{n,m}(\pi)$ be its order distribution.

Theorem 1. *For $n \geq 3$ and $m > n/2$, every permutation satisfies*

$$1 - \frac{n(n-1)(n-2)}{6m^2} \leq n!P_{n,m}(\pi) \leq \exp\left(\frac{n(n-1)(n-2)}{12(m^2 - n^2/4)}\right).$$

Consequently, for $0 < \varepsilon \leq 1$, the choice

$$m = \left\lceil \sqrt{\frac{n^2}{4} + \frac{n(n-1)(n-2)}{6\varepsilon}} \right\rceil$$

gives $|n!P_{n,m}(\pi) - 1| \leq \varepsilon$ for every π . Thus $O(n^{3/2}\varepsilon^{-1/2})$ faces per die suffice. For $n = 2$, the word 1221 is already exactly fair.

Proof. Represent each chosen face by a uniform level in $[m]$ and a uniform choice of one of the two occurrences within that level. A block restricted to any two letters is pairwise fair. Conditional on every occupied level containing at most two dice, the output permutation is uniform: the allocation of dice to levels is invariant under relabelling, and the order of each two-die cell is a fair independent binary choice. A union bound over triples therefore gives

$$P_{n,m}(\pi) \geq \frac{1}{n!} \Pr(\text{no level contains three dice}) \geq \frac{1}{n!} \left(1 - \binom{n}{3} m^{-2}\right).$$

For the upper bound, a prescribed order of $r \geq 1$ different letters has at most two occurrences as a subsequence of B_n . Indeed, such an order must increase and then decrease, and the only possible ambiguity is which of the two occurrences of its largest letter is selected. Its conditional probability within one level is therefore at most 2^{1-r} . For any fixed full permutation, distributing its consecutive segments among the m levels gives

$$P_{n,m}(\pi) \leq m^{-n[z^n]} \left(1 + \sum_{r \geq 1} 2^{1-r} z^r\right)^m = m^{-n[z^n]} \left(\frac{1 + z/2}{1 - z/2}\right)^m.$$

In fact equality holds for the increasing permutation, because every consecutive segment of that permutation is admissible. After the substitution $t = mz$, the right side, multiplied by $n!$, is

$$n![t^n] \exp \left(t + \sum_{j \geq 1} \frac{t^{2j+1}}{(2j+1)2^{2j}m^{2j}} \right).$$

When the exponential is expanded, a choice of terms of total degree d contributes the falling factorial $n^{\underline{d}}$. The elementary inequality

$$n^{\underline{d_1 + \dots + d_a}} \leq \prod_{b=1}^a n^{\underline{d_b}} \quad (d_b \geq 0, \sum_b d_b \leq n)$$

allows the contributions of distinct non-linear terms to be separated. Terms whose total degree exceeds n only increase the resulting bound. It follows that

$$n!P_{n,m}(\pi) \leq \exp \left(\sum_{1 \leq j \leq (n-1)/2} \frac{n^{2j+1}}{(2j+1)2^{2j}m^{2j}} \right).$$

Since $n^{\underline{2j+1}} \leq n^3 n^{2j-2}$ and $2j+1 \geq 3$, this is at most

$$\exp \left(\frac{n^3}{12m^2} \sum_{j \geq 1} \left(\frac{n^2}{4m^2} \right)^{j-1} \right),$$

which is the asserted upper bound. For the specified m , its exponent is at most $\varepsilon/2$, and $e^{\varepsilon/2} \leq 1 + \varepsilon$ on $0 \leq \varepsilon \leq 1$. The lower error is at most ε . $\square$

A finite obstruction for this family. The coefficient calculation for the increasing permutation is exact and has nonnegative terms. Keeping the first nonconstant term gives

$$n!P_{n,m}(12 \cdots n) - 1 \geq \frac{n(n-1)(n-2)}{12m^2}.$$

Thus relative error at most ε in this family requires $m^2 \geq n(n-1)(n-2)/(12\varepsilon)$. The sufficient bound above is within an asymptotic factor $\sqrt{2}$ in the number of faces when $n/\varepsilon \rightarrow \infty$. This also shows directly, without an asymptotic shuffle theorem, the sharp $n^{3/2}/\sqrt{\varepsilon}$ scale for this family.

Verification. The accompanying `verify_shelf.py` computes subsequence counts using integer arithmetic, checks every permutation for small n, m , compares the maximum with the generating-function formula, and checks the stated bounds. The proof, rather than numerical agreement, establishes the arbitrary-size claims.